SOVEREIGN CLOUD DAYS 2026 · WORKSHOP STAGE

# Agentic AI meets Sovereign Cloud

openEduSuite on SCS-verified legacy bare-metal hardware

Sun, 25.10.2026 · 11:15 · 60 min incl. Q&A

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MOTIVATION · 3 MIN

# Legacy hardware isn't junk

- Data centers have **decommissioned servers** — often fully functional.
- A sovereign, AI-capable setup runs on **written-off iron**.
- 3 × Huawei RH2288E V2 (~2013) → SCS-verified K3s cluster.

openEduSuite: HFD Digital Sovereignty Ideas Competition — winner 2026-10-06.

WS **Who runs “legacy hardware” in production themselves?**

MOTIVATION · THE PIVOT · 3 MIN

# From OpenStack to bare-metal K3s

**OpenStack / OVN on VMs**

OVN 25.03.1 gateway bug
FIPs dead, DHCP broke

“Reboot destroys the fix”

Pivot

**Bare-metal K3s (SCS-verified)**

3 nodes · Flannel · MetalLB · Ceph
6/6 KaaS standards PASS · 444/444 CNCF

fewer layers, fewer failure points

The VM/SDN abstraction was too heavy for 3 nodes — bare metal is leaner.

HANDS-ON · PI.DEV INSTALL · 2 MIN

# Install pi

One command is enough — the free tier needs no API key.

```
curl -fsSL https://pi.dev/install.sh | sh
```

- Alternatively: `npm i -g @earendil-works/pi-coding-agent` (Node 22.19+)
- Verify: `pi --version`
- Start: `cd <projekt> && pi`

HANDS-ON · CHOOSING A MODEL · 3 MIN

# Start for free — big pickle

- `/login` → provider **opencode**
- `/model` → **big pickle** (large, free)
- Or free tier: `GROQ_API_KEY` / `GOOGLE_API_KEY`

> The “free big pickle” is enough for agentic experiments — no cost risk.

HANDS-ON · PROMPT IDEAS · 6 MIN

# Starter prompts to try

- “Explain SCS-0217 Pod-Security in 3 sentences.”
- “Write a default-deny NetworkPolicy for the `llm` namespace.”
- “PVC on ceph-rbd, 3× replication — show the manifest.”
- “Ceph is HEALTH_WARN — give 3 first debugging steps.”
- “Outline: student question → retrieval → Ollama CPU → answer.”

WS **Who installed pi in 5 min? → walk around, help.**

CLUSTER · HARDWARE · 3 MIN

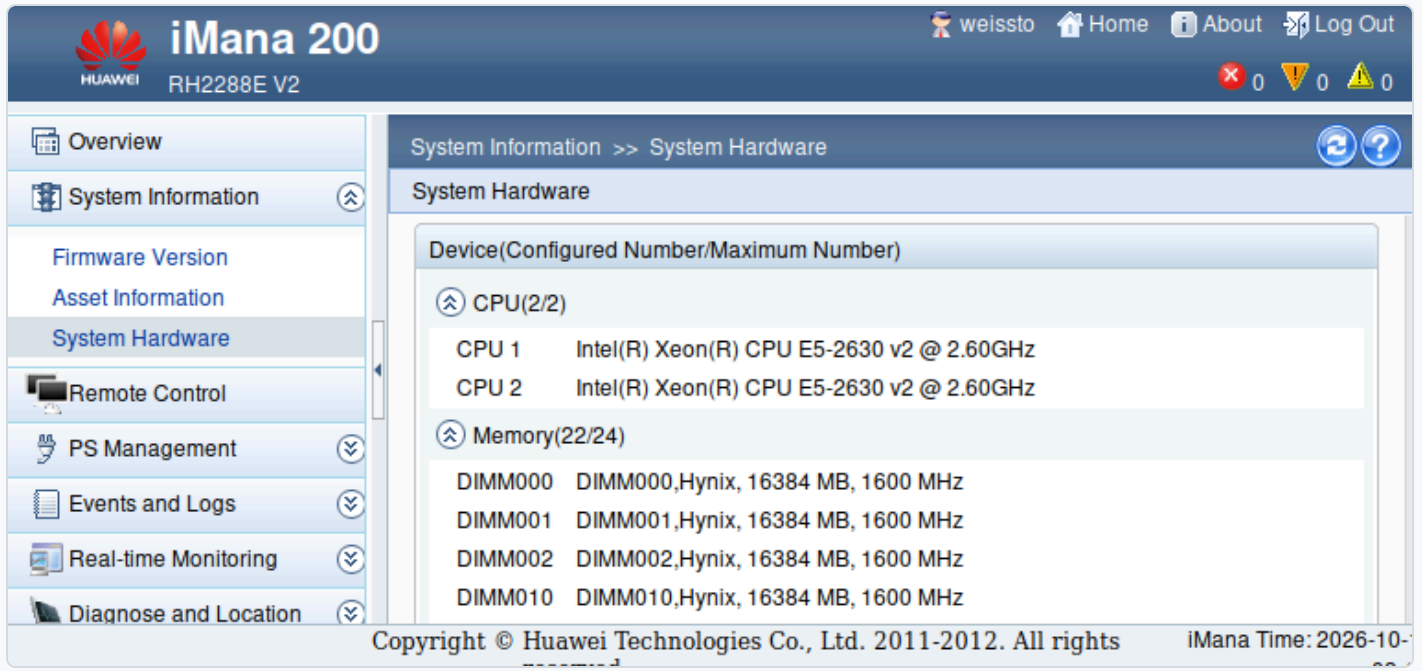
# 3 × Huawei RH2288E V2 — no GPU

**CPU**
2× Xeon E5-2600 v2 (Ivy Bridge-EP) · 6–12 cores

**RAM**
24× DDR3 ECC RDIMM · max 768 GB

**STORAGE**
6× 512 GB SSD · Ceph 8 OSDs · Repl. 3

**NETWORK**
2× Intel X540 10 GbE · no GPU

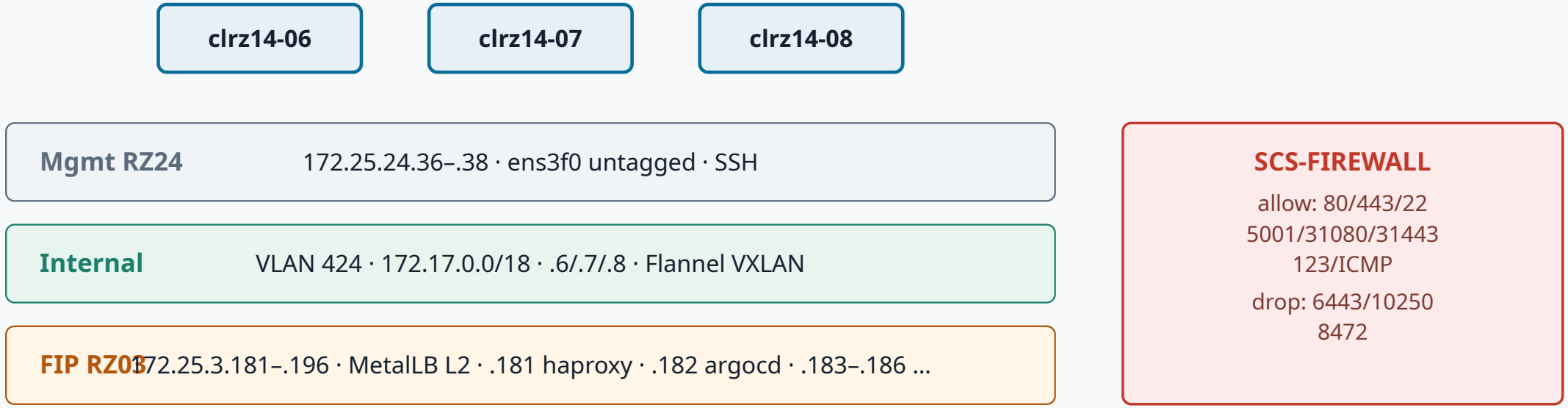



iBMC iMana — hardware view (clrz14-06/07/08)

CPU model / RAM size: readable only via iBMC/BIOS → [verify]

CLUSTER · NETWORK · 3 MIN

# Three networks, one firewall




ens3f0: mgmt + FIP (VLAN 3) · ens6f0: internal (VLAN 424) · iBMC out-of-band

# The stack — bare-metal in production

openEduSuite Workloads (edu · staff · students · sme · llm)

ArgoCD (GitOps) · K8up/Restic (backup → SeaweedFS S3)

Ceph 18.2.8 Reef — 8 OSDs, RBD + CephFS, Repl. 3

Flannel VXLAN · MetalLB L2 · HAProxy Ingress

K3s v1.36.3+k3s1 · Single-server · embedded datastore

Ubuntu 24.04 LTS · containerd 2.3.3 · runc 1.4.3

3 × Huawei RH2288E V2 (bare metal)

CLUSTER · SCS-VERIFIED · 2 MIN

# 6/6 KaaS standards PASS — measurable

**Compliance double-check — 13/13 PASS, overall readiness 1**

| SCS check | Status | Independent evidence |
|---|---|---|
| SCS-0210 K8s version | ✅ | v1.36.3+k3s1 (within v1.36.x policy window) |
| SCS-0211 Storage | ✅ | default SC = ceph-rbd |
| SCS-0214 Nodes | ✅ | 3 nodes in rz03 (clrz14-06/07/08, zones a/b/c), all Ready |
| SCS-0217 Pod security | ✅ | 8 namespaces enforce=baseline |
| SCS-0219 Networking | ✅ | 36 policies, k3s controller, sig-network conformance passed |
| SCS-0201 Conformance | ✅ | 444/444 verified 2026-08-23 (within 30 d) |
| Ceph health / OSDs | ✅ | ceph_health_status 0, OSDs up |
| API server / pods | ✅ | /readyz ok, 100% pods ready |

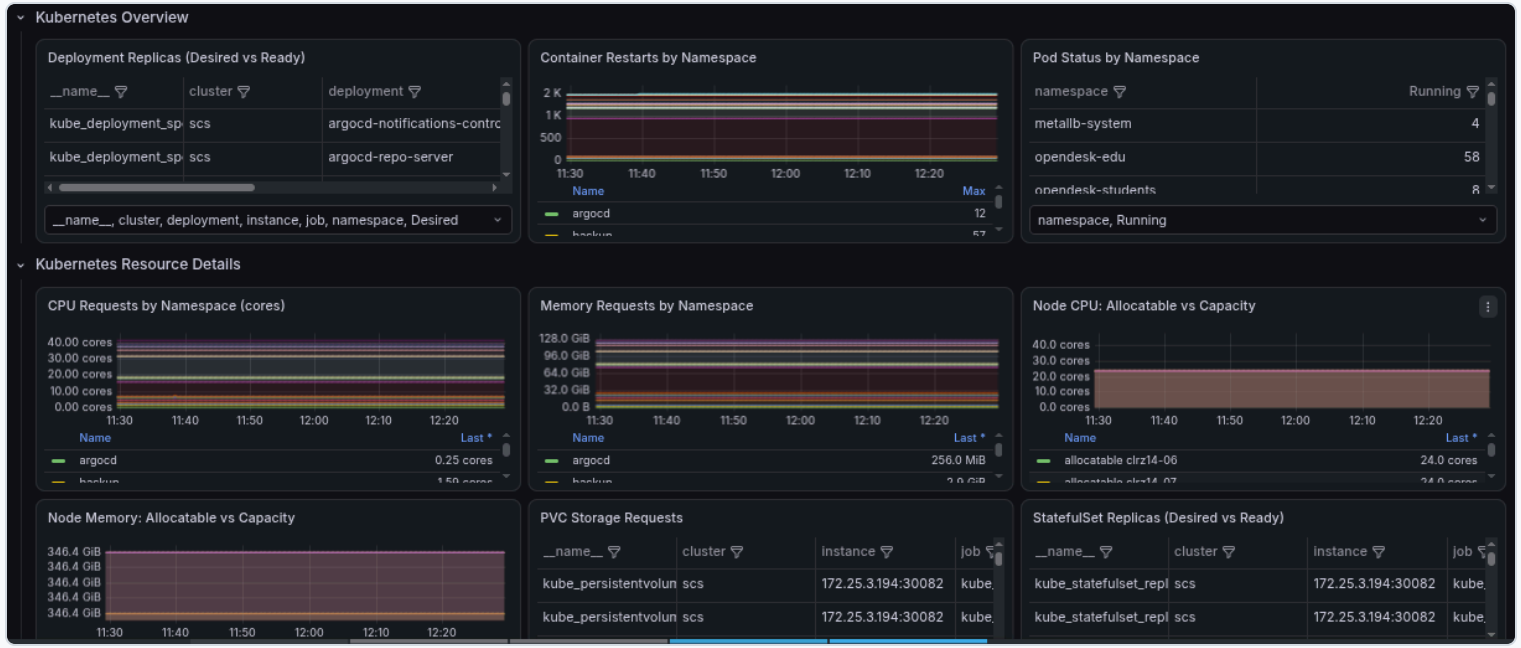
SCS KaaS compliance check (2026-08-23)

- **SCS-0201** CNCF 444/444 · **0210** v1.36.3 · **0211** ceph-rbd
- **0214** 3 nodes · **0217** Pod-Security · **0219** Networking

> Open: OIDC as 2nd auth · Pod-Security 17/22 NS · Loki/Trivy CI · Ceph at rest · GDPR clearance.

CLUSTER · OPERATIONS · 2 MIN

# What we learned in operation




Grafana K3s dashboard (7 dashboards, predictive-agent)

- Stale digests, default-deny NetworkPolicies, OIDC migration — all resolved and documented
- Updates: Ceph-first, rolling, rollback script · backup: K8up → SeaweedFS S3

OPENEDUSUITE · WORKLOAD · 4 MIN

# Sovereign education platform, multi-tenant

TENANTS

edu · staff · students · sme · llm · predictive-agent

AUTH

Keycloak / OIDC · OpenLDAP (.186)

EMAIL

Postfix (.185) · Stalwart (.183) · DKIMpy (.184)

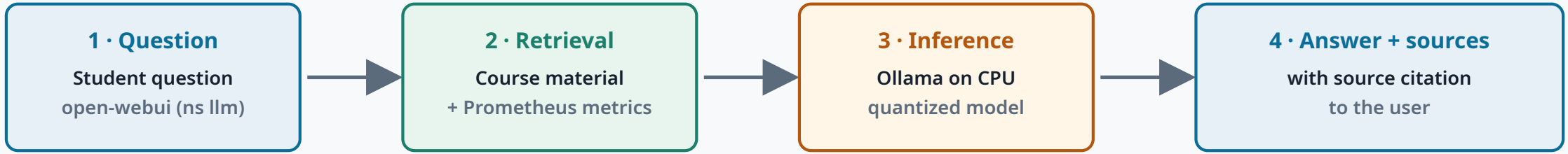
COLLABORATION

Nextcloud · XWiki · Matrix · SOGo6 · OpenProject

Images: our own Nix builds on GHCR → local Zot registry. `opendesk-sme` still Degraded (stale digests).

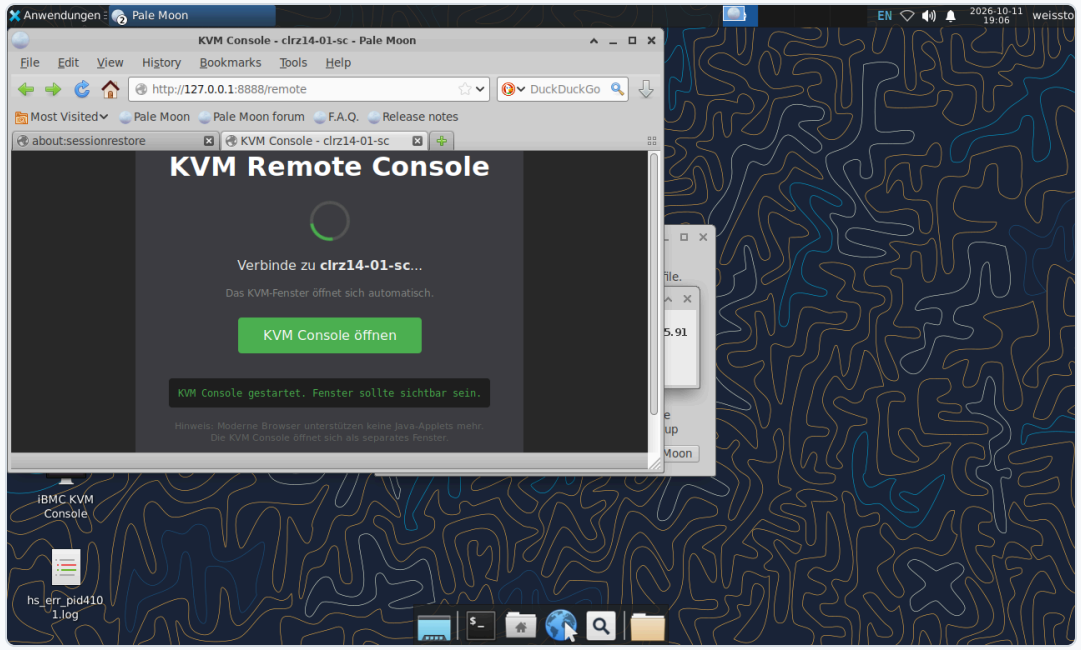
OPENEDUSUITE · AGENTIC WORKFLOW · 4 MIN

# Learning answer in 4 steps




**predictive-agent v8.11-nix:** Risk scoring (Kalman + Markov) for 62 pods — proactive alerting.

# Cluster, login, agent trace




Live iBMC KVM console — server console via Pale Moon + Java Web Start

- Dashboard: 3 nodes · Ceph HEALTH OK (8/8) · SCS board
- Login via Keycloak/OIDC → agentic workflow

> **Fallback:** Screencast + architecture diagram in case the demo stalls.

LESSONS · 3 MIN

# Five takeaways

- **Start small, learn fast** — OpenStack didn't fit the scale.
- **Bare metal instead of a VM layer** — less abstraction, more control.
- **Constraints are features** — quantization beats buying GPUs.
- **“Verified” convinces** — SCS tests open doors, with honest gap reporting.
- **Document like a madman** — runbooks, DPIAs, checklists.

Q&A · 10 MIN

# Questions & discussion

- Architecture, rebuilding it, SCS/openEduSuite licensing topics
- Hands-on: pi.dev installation and prompts from the workshop
- Contact: tobias-weiss.org · graphwiz.ai · ki-kompetenz-training.org

**Backup slides:** architecture diagram · agent trace · screencast · cluster fact sheet.